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B737 FLEET MAGAZINE



An exclusive fleet magazine for EAL 737 pilots.

Fleet News • Deeper look into B737 incidents and accidents • Breakthroughs in flight • Notices •
System reviews

NOTE

This magazine is for information purposes only. **USE COMPANY AND APPROVED MANUALS AS PRIMARY REFERENCE.**

Thank you for taking time to read The EAL 737 Fleet MAGAZINE. All readers are encouraged to submit their questions and suggestions, regarding this magazine to israelya@ethiopianairlines.com

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June 2026



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MESSAGE FROM CHIEF PILOT

Dear Esteemed B737 Pilots,
Warm greetings to you all,

I would like to share an update on the various activities and initiatives within our B737 fleet.

The world is currently facing unprecedented challenges, from climate change driving El Niño to the war in the Middle East reshaping the global economy. As a transnational company, Ethiopian Airlines is inevitably affected, and specifically our **B737 fleet has felt the impact, most of the cancelled flights being 737 flights.** The war in the Middle East has driven fuel prices higher worldwide, while many flights to the region were discontinued.

As a result, our operating costs, as an airline, have risen sharply, with **fuel now accounting for 58% of total costs, compared to the previous international average of 15-30%.** This is unsustainable, and I urge you to continue coming up with innovative ideas to cut costs without compromising safety.

I am proud to highlight the achievements you have already accomplished. Through the senior captain program, you **enabled one B737 previously restricted to fly ETOPS routes to begin flying ETOPS segments,** delivering significant financial benefits by opening new destinations. You also completed the painstaking paperwork to implement the APU on demand procedure on the B737 MAX, saving millions annually.

Captain Bereket G and the maintenance focal team identified the root cause of repeated brake failures on the MAX, **reducing incidents to nearly zero.** **Captain Amanuel Caweley** observed improper cleaning practices that affected carbon brakes and coordinated with maintenance to release a **Maintenance Service Letter** requiring proper protection during cleaning. These are just a few of the cost saving and safety enhancing initiatives achieved in the past year.

Ongoing projects are equally impressive. **The Point Merge System (PMS),** being finalized with Civil Aviation, will save millions yearly by improving ATC efficiency. **The Single Engine Taxi Out (SETO)** procedure for the B737 MAX has completed its trial phase, training is underway, and we await manufacturer feedback before SOP release. This will reduce delays during peak departures. **The outstation coordination team led by Captain Tsega K has launched a Telegram BOT to support pilots, while Captain Hermon and his team continue to coordinate with scheduling to improve working conditions.** None of these achievements would have been possible without the persistent assistance of Deputy Chief Pilots **Captain Dawit Feyisa and Captain Mardokay.**

Continued on next page ...

Looking ahead, the introduction of a Document Management System (DMS) will streamline compliance and eliminate duplication, raising our documentation standards to a worldclass level.

I must emphasize the importance of safety. Our motto remains: **“If you see something, say something.”** Do not ignore hazardous situations—report them. As the rainy season intensifies, be vigilant: runways become slippery, visibility decreases, and thunderstorms grow stronger. Always know your limits, improve your knowledge, and strengthen your resilience as professional pilots. **You are the last line of defense for safety.**

The challenges we face are real: rising costs, harsher environmental conditions, stronger turbulence, and tighter regulations. Yet as the saying goes, “When the going gets tough, the tough get going.” We become tougher by staying safety-minded, innovative, and united in team spirit. First Officers, I encourage you to continue developing yourselves through the monthly **FO development program led by Captain Jegol.**

Finally, I extend heartfelt congratulations to our newlyweds and new parents, and deepest condolences to those who have lost loved ones. As our airline expands with multiple deliveries expected in the coming months, let us tighten our belts and move forward together.

**Blue Skies and Safe Flight,
Capt. NATHAN ELIAS
Chief Pilot - B737&B767**



WET/CONTAMINATED RWY

GENERAL CONSIDERATIONS

INTRODUCTION

Takeoff and landing are the most demanding phases of flight, requiring precise aircraft handling, sound judgment, and accurate performance planning. When runway conditions deteriorate due to rain, standing water, slush, snow, or ice, the margin for error decreases significantly. Reduced runway friction affects acceleration, braking effectiveness, and directional control, increasing the likelihood of runway excursions if appropriate precautions are not taken.

Although modern Boeing 737 aircraft are equipped with advanced systems such as anti-skid protection, autobrakes, speed-brakes, and thrust reversers, these systems cannot overcome the physical limitations imposed by a contaminated runway. Flight crews must therefore understand how runway conditions influence aircraft performance and apply the appropriate operational techniques to maintain safe margins during both takeoff and landing.

Wet vs. Contaminated Runways

Although the terms are often used interchangeably, they have different operational meanings.

Wet Runway

- Water depth ≤ 3 mm (1/8 in.)
- Surface appears reflective
- Moderate increase in landing distance
- Higher hydroplaning risk at high speed

Contaminated Runway

- Standing water, slush, snow, or ice affecting more than 25% of the runway
- Contaminants significantly reduce braking and directional control
- Reduced braking and Significant increase in takeoff and landing distances

Safety Reminder

Never assume a runway is simply "wet". Always review the latest ATIS, Runway Condition Report (RCR), Runway Condition Code (RwyCC), and pilot braking action reports before conducting takeoff or landing performance calculations.

Hydroplaning

One of the greatest hazards associated with wet runway operations is hydroplaning, which occurs when a layer of water prevents the aircraft tire from making direct contact with the runway surface. When this happens, braking effectiveness and directional control are significantly reduced.

Types of Hydroplaning

1. Dynamic Hydroplaning- Occurs when the aircraft travels over standing water at a speed where the tire cannot displace the water quickly enough.

2. Viscous Hydroplaning- Occurs on smooth runways with a thin film of water.

3. Reverted Rubber Hydroplaning- Occurs when a locked tire generates intense heat, converting water beneath the tire into steam and allowing the tire to ride on this vapor layer.

Performance Effects During Takeoff

Wet and contaminated runways affect takeoff performance by increasing rolling resistance and reducing tire traction.

Typical effects include:

- Increased takeoff distance.
- Reduced aircraft acceleration.
- Increased accelerate-stop distance.
- Lower allowable takeoff weight.
- Greater dependence on accurate V-speed calculations.

Soft contaminants such as standing water and slush also create **displacement** and **impingement drag**, further reducing acceleration.

Performance Effects During Landing

Landing performance is generally more affected than takeoff performance because stopping the aircraft relies heavily on braking action.

Reduced runway friction may result in:

- Longer landing distance.
- Delayed deceleration.
- Reduced directional control.
- Increased hydroplaning risk.
- Greater dependence on reverse thrust and spoilers.

Golden Rules for Wet and Contaminated Runway Operations

Before Takeoff

- Obtain the latest ATIS and Runway Condition Report (RCR).
- Verify the Runway Condition Code (RwyCC).
- Confirm aircraft weight and performance calculations.
- Review rejected takeoff procedures.
- Respect crosswind limitations.

Before Landing

- Conduct a thorough approach briefing.
- Review braking action reports.
- Verify landing performance calculations.
- Plan autobrake and reverse thrust usage.
- Always have a go-around strategy.

After Touchdown

- Touch down within the touchdown zone.
- Deploy speedbrakes immediately.
- Apply reverse thrust promptly.
- Maintain runway centerline.
- Monitor deceleration continuously.
- Transition smoothly to manual braking when required.

Crosswind Considerations

Crosswinds significantly increase the workload during wet and contaminated runway operations because reduced runway friction decreases lateral tire grip.

Flight crews should:

- Apply smooth rudder and control wheel inputs.
- Maintain runway centerline throughout the rollout.
- Avoid aggressive nosewheel steering corrections.
- Respect company crosswind limitations.
- Consider delaying or diverting if operational limits are exceeded.

Maintaining directional control is just as important as reducing speed.

**DO NOT HESITATE TO GO
AROUND IF THE APPROACH IS
UNSTABLE !**



RNP APPROACHES

INTRODUCTION

In the evolution of modern aviation, we have moved beyond the era of navigating by ground-based "lighthouses." Traditional navigation relied on station-referenced flying, where accuracy was a byproduct of proximity to a VOR or NDB.

The advent of Performance-Based Navigation (PBN) flipped this script, placing the burden of accuracy and monitoring entirely on the aircraft's onboard architecture.

The ICAO PBN Manual (Doc 9613) identifies two kinds of PBN approach: RNP APCH and RNP AR APCH.

An RNP (Required Navigation Performance): approach is a type of Performance-Based Navigation (PBN) that uses onboard avionics and satellite positioning to guide aircraft along highly precise, 3D-defined paths without relying on traditional ground-based navigational aids like ILS or VOR.

The defining feature of an RNP approach is its onboard performance monitoring and alerting capability the aircraft's Flight Management System (FMS) continuously calculates how accurate its position is (Actual Navigation Performance) and instantly alerts the flight crew if it falls outside the acceptable safety limits.

RNP approaches drastically improve safety and airport accessibility. They offer several distinct features, segments, and operational requirements that define how they are flown.

RNP AR APCH (Authorization Required): Designed for highly challenging environments (e.g., airports surrounded by mountains or deep valleys). Because these procedures can feature extremely tight accuracy bounds (e.g., down to RNP 0.1) and complex turns right up to the runway threshold, they require special regulatory authorization for both the aircraft and the crew (similar to ILS CAT II/III).

These are usually published as RNAV (RNP).

Key Components of an RNP Approach

- **Precision and Accuracy:** The "RNP value" indicates the required accuracy in nautical miles (NM) that the aircraft must maintain for at least 95% of the flight time. For example, during an RNP 0.3 approach, the aircraft must remain within 0.3 NM of the centerline.
- **Curved Flight Paths (RF Legs):** Unlike traditional straight-in approaches, many RNP procedures utilize **Radius-to-Fix (RF) legs**. These are constant-radius curved tracks that allow aircraft to snake through mountainous terrain, avoid noise-sensitive areas, or maneuver around complex airspace.
- **Vertical Guidance:** RNP approaches provide 3D guidance (both lateral and vertical). This is achieved either geometrically via a Satellite-Based Augmentation System (SBAS, like WAAS in the US or EGNOS in Europe) to **LPV minima**, or by combining GNSS with Barometric VNAV to **LNAV/VNAV minima**.

Approach Segments

An RNP approach consists of four distinct phases:

- 1. Initial Segment:** Aligns the aircraft from the en-route structure to the intermediate fix. The required lateral accuracy is typically RNP 1.0.
- 2. Intermediate Segment:** Prepares the aircraft for the final descent.
- 3. Final Approach Segment (FAS):** The final descent path to the runway. During this segment, the RNP value usually scales down tightly, often to RNP 0.3 or even RNP 0.1, ensuring the aircraft is perfectly aligned with the runway centerline.

4. Missed Approach: The procedure followed if the aircraft cannot land. It typically uses an RNP 1.0 or RNP 0.3 requirement to safely guide the aircraft away from obstacles.

Operational Requirements

To fly a standard RNP approach, the following are necessary:

- An IFR-approved GNSS/GPS sensor providing the primary positioning data.
- An up-to-date **navigation database** loaded in the Flight Management System (FMS).
- An autopilot or flight director capable of tracking the 3D paths, particularly if flying RF legs or low-minimum AR approaches.

RNP VS RNP AR

Compared to standard RNP approach procedures, the RNP AR approach procedures are characterized by:

- RNP values ≤ 0.3 NM
- Curved flight path before and after the Final Approach Fix (FAF) or Final Approach Point.
- Protection areas laterally limited to $2 \times$ RNP value without any additional buffer.
- RNP AR operations may include missed approach procedures and instrument departures with reduced RNP (<1 NM). The RNP AR operations are accessible to aircraft and operators complying with specific airworthiness and operational requirements.

- RNP AR APCH have stringent equipment and pilot training standards and require special authorization to fly.
- Scalability and RF turn capabilities are mandatory in RNP AR APCH eligibility.
- RNP AR APCH vertical navigation performance is based upon barometric VNAV or SBAS.
- RNP AR is intended to provide specific benefits at specific locations. It is not intended for every operator or aircraft.
- RNP AR capability requires specific aircraft performance, design, operational processes, training, and specific procedure design criteria to achieve the required target level of safety.

Benefits of RNP AR Approach

Some of the benefits of **RNP AR APCH** are:

- Allows procedures to be implemented in challenging obstacle environments where conventional obstacle protection surfaces limit the possibility of implementation.
- RF (Radius-to-Fix) legs are used to advantage.
- Improves runway (RWY) alignment.
- The approach is APV (Approach Procedure with Vertical Guidance).
- Enables lower minimums.
- Missed approach guidance provides improved safety and operating minima.
- Improves efficiency and reduces emissions.

Operating Procedures

Preflight Considerations

1. Minimum Equipment List (MEL)

should be checked to address the equipment requirements for RNP AR instrument procedures. The required equipment may depend on the intended navigation accuracy and whether the missed approach requires an RNP less than 1.0. For example: GNSS and autopilot are typically required for high navigation accuracy. Dual equipment is typically required for approaches using minima less than RNP 0.3 and/or where the missed approach requires an RNP less than 1.0.

2. Autopilot and Flight Director (FD)

RNP AR APCH procedures with: lateral navigation accuracy less than RNP 0.3, or RF legs, require the use of an autopilot or flight director driven by the RNP system in all cases. The autopilot/flight director must be capable of tracking both the lateral and vertical paths defined by the procedure.

3. Navigation Database Currency

During system initialization, pilots must verify that the navigation database is current. The navigation database is expected to remain current throughout the flight.

In-Flight Considerations

Modification of the Flight Plan-Pilots are not authorized to fly a published RNP AR APCH unless: It is retrievable by procedure name from the aircraft navigation database. It conforms exactly to the published procedure.

The lateral path must not be modified except for:

- Accepting a clearance direct to a fix before the FAF that does not immediately precede an RF leg.
- Changing altitude and/or speed constraints on the initial, intermediate, or missed approach segments (e.g., cold temperature corrections or ATC instructions).

Required List of Equipment Pilots must have the required equipment list or approved alternate procedures to address equipment failures that would prohibit continuation of an RNP AR APCH.

RNP Management Pilots must ensure the navigation system uses the correct navigation accuracy throughout the approach. When multiple lines of minima exist: Verify the desired navigation accuracy has been selected. If the system does not automatically select the required accuracy from the navigation database, manually select the smallest RNP required for the approach and missed approach before beginning the procedure.

GNSS Updating All RNP AR procedures require GNSS updating. Pilots must verify GNSS updating is available before commencing the procedure. If GNSS updating is lost and the system cannot continue meeting performance requirements, the approach must be discontinued unless the required visual references are in sight.

Radio Updating Initiation of RNP AR APCH procedures is based on GNSS availability. DME/DME updating may be used as a reversionary mode when permitted and when required navigation accuracy is maintained. VOR updating is not authorized. Pilots must follow company procedures regarding inhibition of specific navigation facilities.

Track Deviation Monitoring Pilots must use: Lateral deviation indicators and/or Flight Director in lateral navigation mode. Aircraft must have appropriate CDI scaling for each segment of the procedure. Normal operations require cross-track error to remain within approximately $\pm\frac{1}{2}$ of the navigation accuracy for the procedure segment. Briefovershoots or undershoots immediately after turns, up to one times the navigation accuracy of the procedure segment, are acceptable.

Temperature Compensation For aircraft equipped with approved temperature compensation: Approved operating procedures may allow pilots to disregard published temperature limits. Pilots must receive operator-approved training. System temperature compensation applies only to Baro-VNAV guidance. It does not replace pilot cold-temperature altitude corrections for minimum altitudes or Decision Altitude (DA).

Altimeter Setting RNP AR APCH procedures use barometric altitude information. Pilots must: Set the current local QNH before the FAF. Not use remote altimeter settings. Altimeter Cross-Check Both pilots' altimeters must agree within: ± 30 m (± 100 ft) Before the FAF (but not before the IAF). If the cross-check fails, the approach must not be continued.

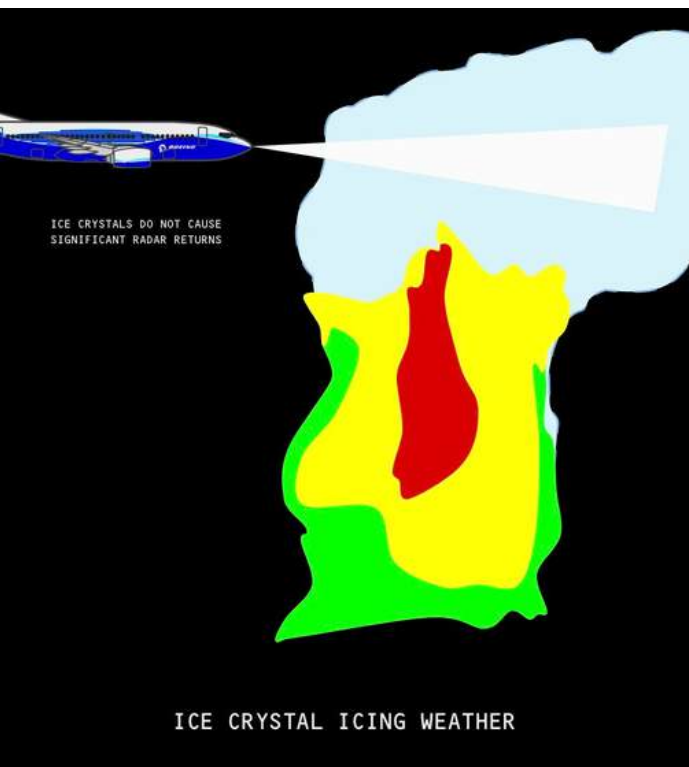
VNAV Altitude Transitions The aircraft's Barometric VNAV system provides fly-by vertical guidance. As a result, the aircraft may begin intercepting the vertical path before the FAF. Small vertical displacements occurring near vertical constraints (such as the FAF) are considered operationally acceptable provided the required obstacle clearance and procedure design criteria are maintained.

Vertical Deviation Vertical deviation should remain within: 22 m (75 ft) during the Final Approach Segment (FAS). Temporary excursions greater than 75 ft above the vertical path may occur during configuration changes or energy management. Pilots must execute a missed approach if: Lateral deviation exceeds $1 \times \text{RNP}$, or Vertical deviation exceeds ± 75 ft, unless the required visual references have been acquired.

Go-Around or Missed Approach Where possible, the missed approach will require a navigation accuracy of RNP 1.0. The missed approach portion of these procedures is similar to the missed approach of a standard RNP APCH. Where necessary, navigation accuracy less than RNP 1.0 will be used during the missed approach.

Important Considerations

- When initiating an early go-around, the pilot must remain on the published track unless ATC has issued an alternative clearance.
- Pilots should also be aware that **RF (Radius-to-Fix) legs** are designed for a maximum ground speed. Initiating an early go-around at speeds higher than those assumed during procedure design may cause the aircraft to diverge from the intended turn and require pilot intervention to maintain the correct flight path.



ICE CRYSTAL ICING

The Invisible High-Altitude Threat

1.0 INTRODUCTION

Ice Crystal Icing (ICI) is a unique form of engine icing that differs significantly from conventional airframe icing. While most pilots associate icing with temperatures near freezing and visible ice accumulation on the aircraft, ICI occurs at high altitudes in extremely cold conditions, often when the aircraft itself remains completely free of ice.

Historically, ice crystals at high altitude were not considered a threat to airplanes because they did not lead to airframe icing. However, the industry has identified a condition in which solid ice particles can affect both engines and air data probes.

ICI occurs at cruise altitudes, typically near deep convective systems, where outside air temperatures can be below -40°C . Unlike supercooled water droplets, these ice crystals do not readily adhere to the airframe or engine inlet. Instead, they pass into the engine, where they partially melt in warmer compressor stages before refreezing on colder internal components, leading to ice accretion inside the engine.

The consequences can be significant: **engine vibration, compressor stalls, thrust rollback, or even engine flameout.** Since the airframe often remains free of ice, the threat can go unnoticed until engine indications begin to change.

Despite pilot avoidance of weather radar returns, engine power losses have occurred.

Avoidance of ice crystals is a challenge because they: are not easily identified visually, often do not produce noteworthy weather radar returns, and can occur in high concentration of ice crystals even when avoiding thunderstorms by more than the standard 20-mile guidance

2.0 INDICATIONS OF ICI

The following are threats associated with ICI:

- In Clouds at High Altitude: All pilots report being in clouds when the ice crystal-induced engine power loss encounters occur.
- No Weather Radar Returns at Flight Level: ICI particles are very small, about the size of baking flour. Even in high concentrations, they are a poor reflector of radar energy.
- Heavy Weather Radar Returns Below Flight Level: Analysis of weather radar encounters of ICI show that they consistently occur when the airplane is in IMC and over-flying an area which would be amber or red on the pilot's weather radar if tilted down. This is because larger/wetter/warmer particles at lower altitude are usually much more reflective to radar energy. Heavy radar returns can be identified by pilots if they manually tilt the radar down to scan below the freezing level.
- Traversing Clouds with Tops at High Altitude: Often these clouds lift condensed water to high altitudes. Sometimes they even penetrate the tropopause. It is in these updraft areas where the highest ice crystal concentrations can be encountered.
- No Airframe Icing: Ice crystals bounce off cold surfaces such as the airframe, which is why airframe icing is usually not noted. These small particles may accumulate in stagnation areas, so it is possible that a small concentration might be noted on the leading edge of the windshield wiper post.
- Appearance of Rain: Another key indicator of ice crystal conditions is the appearance of rain on the windshield at high altitudes. After analyzing pilot reports and flight data it is concluded that high concentrations of ice crystals melt when they contact a heated windshield. This can occur even when the ambient outside air temperature is too cold for liquid to exist. Several interviewed pilots explained that ice crystals sound different than typical rain, producing a "ssshhh" sound.

- **Total Air Temperature (TAT) Near 0°C:** A TAT anomaly is indicated when the TAT suddenly changes to 0°C during climb, descent, or level flight. This is a good indication the airplane has flown through ice crystals. This anomaly is caused by ice crystals building up in the area near the sensor element. Some ice crystals melt, and the sensor measures 0°C. In some cases, TAT has “flat-lined” at 0°C during a descent and may be noticeable to pilots. The TAT anomaly is not a reliable indicator of ICI conditions because it does not occur on all airplanes and it sometimes occurs after the engine encounter.
- **Only Light to Moderate Turbulence:** The tropical thunderstorm complexes that most often create ICI are also less aggressive than other thunderstorms. Pilots often avoid the strongest cores, remaining in the less aggressive anvil portions, resulting in ICI encounters with no unexpected turbulence.
- The appearance of St. Elmo’s fire is associated with ice crystals striking the airframe.
- Increases in temperature and/or humidity, including the smell of ozone, on the flight deck preceding a TAT anomaly are other frequently reported flight deck effects during flight through ICI. These effects were only present while in IMC and are additional clues that ice crystals were present.
- High fan vibrations (above 4 flight deck units) have been reported during flight in ICI. The cause is ice crystals accreting under the fan hub (specifically, the fan blade platform seal). The vibrations resolve once the airplane has exited the convective weather and the accretions either dissolve through sublimation or shed, or melt in above-freezing conditions.

3.0 Key Points for Flight Deck Crews

- Avoid flight through or beneath thunderstorm anvils, even when radar returns appear weak.
- Remember that hazardous ice crystal concentrations may exist well beyond the visible storm, sometimes more than 50 NM downwind.
- Monitor engine parameters closely when operating near deep convection at cruise altitude.
- If Ice Crystal Icing is suspected, follow the applicable QRH/FCOM procedures and exit the affected area as soon as practical.



MAX BRAKE CONCERNS

INTRODUCTION

Recent incidents involving Boeing 737 MAX aircraft and braking systems have drawn renewed attention from airlines.

Over the past year, several 737 MAX flights reported brake-related events during landing or rejected takeoff, including overheated brakes, tire damage, and in some cases smoke or fire near the landing gear.

Modern commercial jets including the 737 MAX typically use carbon composite brake systems. Compared with traditional steel brakes, carbon brakes are lighter and can handle high energy loads. They also generally cool faster, which can help reduce ground turn-around times.

However, these systems still have limits. Excessive heat from long taxi phases, repeated high-energy braking, or other typical operations can push parts beyond their thermal capacity.

If internal temperatures climb too high, air pressure differences and friction can lead to tire bursts, smoke, or even wheel assembly damage.



Effective mitigation of problems concerning brake systems on the 737 MAX depends on a combination of correct pilot technique, maintenance practices, and operational planning.

Here are some key operational practices pilots can use to reduce brake stress on the 737 MAX, while maintaining safe stopping performance under all conditions.

1. Optimized Use of Autobrakes

The autobrake system on the 737 MAX is designed to provide consistent, predictable deceleration while minimizing unnecessary brake pressure.

Best practices include:

- Selecting the lowest autobrake setting that safely meets runway length, surface condition, and performance requirements.
- Avoiding excessive manual brake inputs that override the autobrake unless operationally necessary.
- Allowing the autobrake system to manage brake pressure smoothly rather than applying abrupt pedal inputs.

Using autobrakes correctly reduces peak brake temperatures and distributes braking energy more evenly across the brake assemblies.

2. Effective Use of Reverse Thrust

Reverse thrust is one of the most effective tools for reducing brake workload after touchdown.

Best practices include:

- Apply reverse thrust promptly after main gear touchdown, in accordance with standard operating procedures.
- Using full reverse thrust is the most effective way to reduce brake temperature, especially at high gross weights.

- Gradually reduce reverse thrust as speed decreases, transitioning smoothly to wheel braking.

By maximizing aerodynamic and engine deceleration early in the landing roll, pilots significantly reduce the energy that must be absorbed by the wheel brakes.

3. Smooth Manual Braking Techniques

When manual braking is required, technique matters.

Best practices include:

- Applying brakes progressively rather than aggressively, allowing the anti-skid system to operate efficiently.
- Avoiding rapid on-off brake pedal inputs, which increase heat buildup and wear.
- Maintaining directional control with rudder rather than differential braking whenever possible.

Smooth braking minimizes brake temperature spikes and reduces uneven wear across brake units.

4. Taxi and Ground Operations Discipline

Brake stress does not occur only during landing—taxi operations also contribute significantly to brake wear and heat buildup.

Best practices include:

- Minimize rapid exit taxi speed as much as possible. Lower speed reduces the risk of tire related issues.
- Minimize brake use during taxi by allowing the aircraft to roll at idle thrust where possible.

- Avoid riding the brakes, especially after landing when brake temperatures are already elevated.

Efficient taxi techniques help prevent brake overheating and reduce cooling time before the next departure.

Reducing brake stress on the Boeing 737 MAX is not solely a maintenance issue but a shared operational responsibility between flight crews, dispatchers, and ground personnel.

Through proper autobrake selection, effective use of reverse thrust, smooth braking techniques, disciplined taxi practices, and careful consideration of environmental factors, pilots can significantly reduce brake wear and heat buildup.

These operational techniques not only enhance safety margins but also improve aircraft availability, reduce maintenance costs, and contribute to more reliable day-to-day operations



MAX SETO PROCEDURE

1.0 Introduction

As fuel prices continue to challenge our airline's profitability, operational efficiency has become more critical than ever. Among the strategies FOPs is working to implement is the 737 MAX SINGLE ENGINE TAXI OUT (MAX SETO) procedure, which allows operators to optimize fuel costs while maintaining the highest safety standards.

This article is to familiarize crew members with MAX SETO procedures before implementation is officially announced.

Refer FCOM ENGINE OUT SUPPLEMENTARY PROCEDURE AND BOEING'S FOTB Number 737-17-01 SUBJECT: Engine out taxi.

Do not accomplish SETO for the following conditions;

- ICING conditions exist
- Local airport restriction.
- Taxi surfaces are contaminated
- **MEL restrictions that impact braking &/or steering, Electrical generation, Engine reliability...**
- Slope on Apron/ramp or parking conditions that adversely affect maneuverability in the gate/stand/parking spot.

General considerations when performing SETO

- Do the normal engine start procedure except only start ENGINE NO.2
- EOT is subject to PIC's discretion.
- The F/O should accomplish engine start operation and monitoring while captain remains responsible for taxing.
- Start both engines if there are concerns about the amount of breakaway thrust needed to begin taxi. If necessary, stop the airplane, set the parking brake and start the other engine.

- GTOW LESS THAN 70,000 KG FOR -800S & 62,000 KG FOR -700S & LESS THAN 75,000KG FOR MAX.
- Run the engines for at least 3 minutes before setting takeoff thrust.
- **A minimum of estimated 5 minutes taxi time for 737NG and 6 minutes for 737MAX.**
- **APU is available**
- Engines are not considered cold. (Not more than 4hrs since last Shutdown)
- During taxi tight turns have to be avoided & an energy management technique using higher ground speed should be applied while turning towards the live Engine.
- During the second engine start to manage workload A/c stationary preferable.
- At the discretion of the Captain, SETO procedures should be performed by memory or reviewing the procedure prior to accomplishment.

SETO BEFORE TAXI PROCEDURE

After Engine no. 2 has been started and stabilized:

Call “FLAPS____ SETO BEFORE TAXI PROCEDURE”C

Flap lever..... Set Takeoff Flaps F/O
MFD SYS Switch..... Push F/O
GENERATOR 2 switch..... ON F/O
PROBE HEAT switches..... ON F/O
WING ANTI-ICE switch..... As required F/O
ENGINE ANTI-ICE switches..... As required F/O
PACK switches..... AUTO F/O
ISOLATION VALVE switch..... CLOSE F/O
APU SWITCH..... ON F/O
APU GEN Switch 1..... ON F/O
Power AC transfer bus 1 with APU generator.

APU BLEED air switch..... ON F/O
ENGINE no. 2 ST ART switch.....CONT F/O
Recall..... CHECK F/O
Verify that all system annunciator panel lights illuminate and then extinguish except ELEC, HYD, and AIR COND. Illuminated amber lights are DRIVE 1, ENG 1 HYD pump LOW PRESSURE, DUAL BLEED and GEN OFF BUS blue light.

Flight Controls.....Check C
Call SETO BEFORE TAXI CHECKLIST..... C
SETO BEFORE TAXI CHECKLIST..... Read F/O

SETO BEFORE TAXI CHECKLIST

GENERATOR 2..... ON
PROBE HEAT..... ON
WING ANTI-ICE switch..... As required
ENG NO 2 ANTI-ICE..... As required
ISOLATION VALVE..... CLOSE
No.2 ENGINE ST ART SWITCH CONT
RECALL..... CHECKED
AUTOBAKE..... RTO
ENGINE NO.2 ST ART LEVER..... IDLE DETENT
FLIGHT CONTROLS..... CHECKED
Ground personnel and equipment..... CLEAR

When ready to taxi clear right and left.... Both

SETO ENGINE NO. 1 START PROCEDURE

L PACK switch..... OFF F/O
Lower DU (For NG)..... ENG F/O
Engine no. 1 thrust lever..... CLOSE F/O

Engine no. 1 Start lever operation and parameters monitoring should be accomplished by FO, while the Captain remain responsible for taxiing. Traffic permitting , stop the aircraft to accomplish Engine start.

Call “Start Engine no. 1” C

SETO BEFORE TAKEOFF PROCEDURE

After Engine no. 1 has been started and stabilized

Call “SETO BEFORE T/OFF PROCEDURE”..... C
Primary engine indications OFF F/O
GENERATOR 1 switch..... ON F/O
Left PACK switch..... AUTO F/O
ISOLATION VALVE switch..... AUTO F/O
APU BLEED air switch..... OFF F/O
Engine no. 1 ST ART switch..... CONT F/O

After confirming with the captain
APU switch..... OFF F/O

Call "SETO BEFORE TAKEOFF CHECKLIST" C
SETO BEFORE TAKEOFF CHECKLIST ...Read F/O

SETO BEFORE TAKEOFF CHECKLIST

- GENERATORS..... ON
- ANTI-ICE..... OFF
- ISOLATION VALVE..... AUTO
- ENGINE START SWITCHES..... CONT
- RECALL..... CHECK
- AUTO BRAKE..... RTO
- ENGINE START LEVERS..... IDLE DETENT
- FLAPS _____ GREEN LIGHT
- SATBILIZER TRIM _____ UNITS



MAX WXR RADAR

THE SYMBOLS

1.0 Introduction

Modern airborne weather radar is one of the most important situational awareness tools available to Boeing 737 MAX pilots. Unlike terrain displays or navigation maps, the weather radar allows the crew to "see" precipitation ahead of the aircraft, helping them avoid thunderstorms, turbulence, hail, wind shear, and heavy rainfall. The Boeing 737 MAX uses an advanced Multicar weather radar to detect hazardous weather, improve situational awareness, and support safe decision-making throughout all phases of flight.

Weather Radar Color Scale Summary

Color	Precipitation	Significance
Green	Light	Normally acceptable
Yellow	Moderate	Evaluate carefully
Red	Heavy	Avoid
Magenta	Extreme	Do not penetrate

2.0 Precipitation Colors

These colors represent the rainfall rate or the density of ice crystals within a cloud:

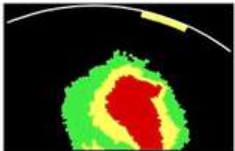
- **Black:** Clear area; no significant return or signal attenuation (shadowing) behind active storm cells. **Black holes within active cells should be avoided.**
 - **Green:** Light precipitation.
 - **Amber (Yellow):** Moderate precipitation.
 - **Red:** Heavy precipitation or dense storm cores.
 - **Magenta:** Very intense precipitation or extreme turbulence
- Turbulence and Hazard Symbols**
- **Magenta Areas:** Highlights areas of severe turbulence (horizontal motion of rain/particles of ≥ 5 meters per second).
 - **Speckled Red / Red & Black Outlines:** Often indicates the presence of hail, high lightning potential, or severe windshear.
 - **Hooks/Fingers:** Visual shapes on the edges of storm cells extending outward may indicate hail shafts or potential tornado development.
 - **Scalloped Edges:** Jagged cell borders suggest significant turbulence that can extend outside the immediate storm cell.

Predictive Windshear

SYMBOL	NAME	MODE	REMARKS
	Predictive windshear symbol (R, B, A)	MAP, MAP CTR, VOR, APP	Displays windshear location and approximate geometric size (width and depth). Amber radials extend from predictive windshear symbol to help identify location of windshear event.
	Windshear annunciation (R, A)	All	Predictive windshear caution active (A). Predictive windshear warning active (R).
	Predictive windshear status annunciation (A)	All	Predictive windshear alerting and display have failed.

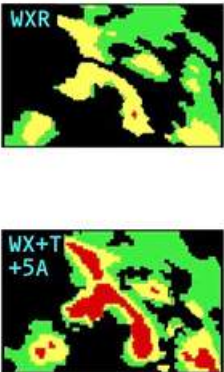
Path Attenuation Compensation (PAC)

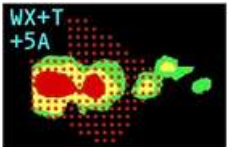
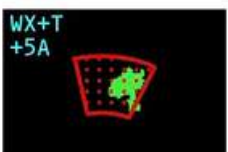
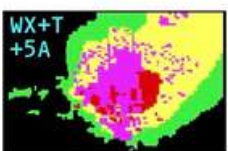
ET-BAT, ET-BAU

	Weather radar Path Attenuation Compensation (A)	MAP, MAP CTR, VOR, APP	Attenuated region behind rainfall (radar shadow) is indicated by amber arc on outer range ring.
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Threat Assessment

ET-BAT, ET-BAU

	Core Threat Assessment (R, A, G, M)	MAP, MAP CTR, VOR, APP	Weather radar colors are adjusted to more accurately reflect the weather threat.
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	Associated Threat Assessment (R)		Lightning or hail probability is indicated by red speckled pattern in black/green/amber areas above 25,000 ft, or green/amber areas below 25,000 ft.
	Predictive OverFlight (R)		Rapidly building cell below flight altitude is indicated by red speckled pattern with red outline. Indication is replaced by actual weather when cell reaches flight altitude.
	Two-level Turbulence (M)		Severe turbulence is displayed in solid magenta. Light-to-moderate turbulence is displayed in speckled magenta.

Weather Hazard Displays

43576 - ET-BAR, ET-BBA - ET-BBD

SYMBOL	NAME	MODE	REMARKS
	Predictive Hail	MAP, MAP CTR, VOR, APP	Conditions in the associated weather cell are conducive to the development of Hail.
	Predictive Lightning	MAP, MAP CTR, VOR, APP	Conditions in the associated weather cell are conducive to the development of Lightning.

	Rain Echo Attenuation Compensation Technique (REACT)	MAP, MAP CTR, VOR, APP	Indicates areas where the radar signal has been attenuated. These areas are shown as magenta arcs superimposed over the reflectivity in the areas where the signal attenuation is significant. These arcs indicate that there could be severe weather in that area, even though only mild or no reflectivity is shown.
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B737 STANDARDIZATION

"When every crew speaks the same operational language, consistency becomes habit—and safety follows naturally."

Enhancing the B737 SOP

The team has been focused on the review and enhancement of the B737 Standard Operating Procedures (SOPs), ensuring they remain aligned with operational experience, manufacturer guidance, and company best practices. Every revision is aimed at making our operations safer, more standardized, and easier to execute.

Introducing the B737 Fleet Quick Reference Guide

The team has also developed the B737 Fleet Quick Reference Guide, a concise one-page reference that consolidates key aircraft limitations, operational differences, and essential reference data for quick access.

The guide is now in its final stage of development and will soon be printed and placed on board every B737 aircraft, ensuring crews have critical information readily available during operations.

This initiative supports improved standardization, enhances operational awareness, and promotes safer and more efficient flight operations.

Coming Soon: Fleet Standardization & Operational Improvement eForm

Development of the **B737 Fleet Standardization & Operational Improvement eForm** is progressing well. The eForm enables flight crews to submit operational observations, SOP improvement proposals, standardization concerns, safety recommendations, and best practices directly to the Standardization Team.

The eForm will soon be available on Logipad, making it easier than ever for crew to contribute valuable feedback from the line.

"Professionalism is doing the right thing every flight—even when no one is watching."

Together, we continue to strengthen our safety culture through standardization, collaboration, and continuous improvement.

Your Feedback Drives Improvement

All B737 pilots are encouraged to make full use of the eForm whenever opportunities for improvement are identified. Many of the best operational improvements begin with a simple observation from the line.

Regularly follow the B737 Standardization Focal Point Telegram channel for the latest standardization notices, SOP updates, operational guidance, and safety information.

Thank You!

